



The Data life cycle in action

Hello, and welcome to this video about effective data reporting!

Being able to share data insights with others is an essential part of data analytics.

After all, an amazing insight to the business question is only valuable when the information is shared!

As a cloud data analyst, you'll likely create a variety of reports to share your analysis and provide summaries for decision-makers.

The data lifecycle — plan, capture, manage, analyze, archive, and destroy — will help you create engaging, data-driven communications.

Now, before we go further, it's important to note that the data lifecycle and the data analysis process are two distinct concepts.

When we think about the **data lifecycle**, this is how the data itself moves and changes throughout its existence.

When we think about the **data analysis process**, this relates to how data analysts interact with the data.

Now that we've covered that!

Let's dive into the data lifecycle.

Let's shift to an on-the-job example with a data analyst, named Kayla.

A **stakeholder** at Kayla's company requests a report of all payments the company received in the last month.

They would like to analyze several invoices to ensure they're not missing any payments.

The stakeholder also wants to see the percentage of payments that were made on time and the payment methods.

To generate this report, Kayla will motion through the entire data cycle.

The first step is the plan stage.

So, Kayla will review the business questions and objectives.

Then, she will determine what fields need to appear on the report.

Kayla considers questions like what filters are needed to extract or separate the data and how far back in time should the data go.

Now, she moves on to the capture stage.

Here, Kayla will implement the plan she's created and assess the metrics she needs to capture.

Then, Kayla will confirm the data is available and then gather the data.

Data can come from a variety of sources.

In the scenario of the payments report, Kayla will need user information and payment details.

Sometimes the data you need doesn't exist.

If that's the case, you may need to work with the product or engineering team to figure out how to fill in the gaps.

You'll also want to establish a reasonable timeline for the data export.

Let's get back to Kayla's process.

The manage stage is all about data management.

In this stage, Kayla considers how to store the data and prioritize security.

Kayla will also determine whether she's working with any PII, or personally identifiable information.

When sensitive data is involved, it's critical to take steps to protect people's privacy.

The report Kayla is generating will likely include user credit card numbers, addresses, and phone numbers.

If she determines this information to be necessary for her project, then she will ensure it's secure.

Otherwise, she will omit the PII at the start and simplify her data security needs.

Kayla can also ensure that the data is accessed on a need-to-know basis so that only people who need the data can access it.

In this phase, Kayla can also perform quality checks to confirm no required values are null or missing.

In these payments report project, let's say she has multiple physical stores collecting payments in addition to an online website.

So, it's important to check that the same fields are using the same kind of data so everything combines seamlessly during table joins.

For example, the field "account number" will need to match to the correct user account.

And both need to match in field types like string values, for example, CUST12345.

Next, it's time to analyze the data.

In this stage, Kayla uses the data to solve problems and support business goals.

She can do this through the collection, transformation, and organization of data in order to draw conclusions, make predictions, and drive informed decision-making.

One of the most exciting parts of the analyze stage is sharing reports of your insights!

Now, Kayla will create visualizations to help others understand the insights.

She will also go back to the security protocol to ensure that she's not sharing any PII with people who should not have access.

Next, in the archive stage, Kayla will determine what needs to be saved.

She may need to save data to share it with additional stakeholders later on.

Or, she might save it to compare with other reports.

Finally, once Kayla determines she doesn't need the data anymore, she moves on into the destroy phase.

Destroying data completes the cycle!

Congratulations, you've just experienced the whole data lifecycle!

You've met your business objectives and kept all of your company's data safe and sound.